

Studying the Benefits of Hybrid Wi-Fi/LiFi Networks for Indoor WLANs

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Abstract—Light Fidelity (LiFi) is a recent wireless technology that offers high data rates, low power consumption, and does not interfere with other radio technologies, such as Wi-Fi. However, its short range and need for a direct line of sight require integration with other wireless technologies for effective use in a wireless local area network (WLAN). In this paper, we study the feasibility and efficiency of Hybrid LiFi/Wi-Fi Networks (HLWNet). Our study aims to determine whether HLWNet is a viable solution for WLANs in terms of throughput and power consumption. We propose an approach to evaluate HLWNet deployments, specifying the optimal number and placement of access points (APs) for each technology to ensure throughput satisfaction for stations (STAs) while minimizing overall network power consumption. Our numerical results indicate that the use of LiFi is not advantageous in all scenarios.

Index Terms—LiFi, Wi-Fi, Hybrid networks, HLWNets, Network efficiency, power consumption, Throughput, Network Deployment, Experimentation

I. INTRODUCTION

LiFi has emerged as a promising technology for short-range wireless communications by transmitting data over visible-light, ultraviolet, and infrared spectrums. IEEE standardization efforts have led to the recent release of the 802.11bb amendment in July 2023, ensuring interoperability between LiFi devices. As a wireless communication technology, LiFi offers significant advantages, as well as an inherent level of security (since light does not penetrate obstacles, it can be easily confined to specific areas), high data rates, and low power consumption. However, its relatively short communication range and the requirement for a direct line of sight make standalone LiFi deployments insufficient for Wireless Local Area Networks (WLANs). Therefore, it must be combined with another technology that offers a longer range (e.g., Wi-Fi or private 5G) to provide seamless coverage and support the mobility of certain stations (STAs), resulting in a hybrid network. The so-called Hybrid LiFi and Wi-Fi Networks (HLWNets), which incorporate both Wi-Fi-based and LiFi-based access points (APs), are viewed as a promising solution for WLANs.

In this paper, we propose an approach to evaluate and compare different deployments of HLWNets. A deployment is defined here as the number of APs, their type (LiFi or Wi-Fi) and their location. This approach allows us to identify suitable deployments for indoor environments that meet the throughput demand while reducing power consumption. This enables to establish the advantages of HLWNets and determine their efficiency in certain real-world use cases.

The study of these deployments must take into account the network needs in terms of load, number of STAs and their

geographical distribution. To fairly compare the deployments with each other, the algorithms defining the assignments between STAs and APs (LiFi or Wi-Fi) must be considered as it greatly impacts the overall performance. We therefore also propose an AP assignment algorithm (APA) that works for HLWNets.

The remainder of the paper is organized as follows: Section II discusses the related work. In Section III, we present experimental results, in which we evaluate the performance (throughput and power consumption) of a commercial LiFi product, and provide measurement values. We introduce our performance and power consumption modeling of HLWNets in Section IV. We present our HLWNets evaluation approach that includes an adequate solution to the APA problem in Section V. Numerical results where we compare different deployments in real-life use cases, while highlighting the conditions under which HLWNets outperform standalone Wi-Fi deployments, are presented in Section VI. Section VII concludes this paper.

II. RELATED WORK

The combination of *Radio Frequency - RF* and optical wireless technologies has been an area of significant interest to researchers over the last years. The readers can refer to the following surveys [1], [2], [3], [4] that present a broad overview of the research in this domain. However, to the best of our knowledge, comparison of different HLWNet deployments to find the best number and positions of APs for each technology has not been addressed in the literature. The state of the art rather tends to focus its efforts on the APA problem for static and dynamic scenarios (assigning a given deployment of the APs).

Access Points Assignment with static (stationary) STAs has been formulated as an optimization problem in [5], [6] and [7], with the goal of enhancing overall throughput. The authors in [6] use a fuzzy logic algorithm to assign STAs to Wi-Fi APs. In [5], authors propose a method that accounts LiFi channel blockages that may prevent AP light from reaching the STAs. The authors of [7] propose two analytical models: a network performance model that considers *Medium Access Control* (MAC) layer, and an energy model for both STAs and APs. Authors in [8] present methods for both centralized and distributed architectures. Their model is upgraded and reused in subsequent studies [9], [10], [11]. In [12], the authors propose an iterative algorithm that incorporates interference between LiFi APs. They also propose an efficient algorithm for the power allocation sub-problem. In [10], the authors present an iterative algorithm based on evolutionary game theory. Their approach considers both the LiFi channel blockage

and the receiver orientation angle. The authors in [13] aim to minimize the network packet delay per STA and compare their solution to the classical received signal strength-based method.

Other papers have attempted to account STAs mobility and handovers. The authors in [14] propose a fuzzy logic algorithm to manage mobility, and they consider the speed of STAs to handle handovers. Authors in [15] focus on the handover management and develop an algorithm to reduce the rate of handovers by predicting the STA trajectory.

Overall, the literature aims to improve the HLWNets performances through the APA problem with a focus on improving network throughput, sometimes combined with other metrics such as handover rate or energy efficiency. However, these papers do not investigate the efficiency of HLWNets by considering different deployments (e.g., deployments with only Wi-Fi APs versus HwLNet). A case in point is that the benefits of HwLNet compared with Wi-Fi only deployments have not been explored. To the best of our knowledge, we are the first to study the performance of HLWNet in various deployments (including a standalone Wi-Fi deployment) and different use cases. Our performance evaluation study includes not only the classical throughput metric, but also the power consumption of the HLWNets assessing different power models.

III. LiFi AP EXPERIMENTATION

Light Fidelity (LiFi), is a wireless communication technology that can use visible light, ultraviolet or infrared spectrum to exchange data. It is bidirectional and can provide high-speed transmission rate (e.g., a transmission rate above 10 Gbps [16]). Since RF and optical technologies do not interfere with each other, LiFi becomes an attractive candidate for sensitive areas where RF products are not allowed (e.g. hospitals or aircraft [17]). We have conducted experiments with a LiFi AP from the market to complete the literature on LiFi performance. This also serves as one of the bases for our models used to evaluate HLWNet deployments.

A. LiFiMax

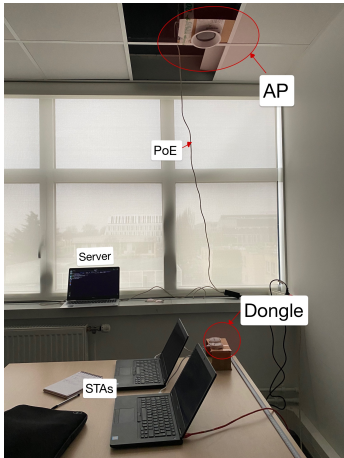


Fig. 1: Setup of the LiFi experimentation.

To carry out our experiments, we select the *LiFiMax* product from the French company *OledComm*. As shown in Figure 1, our experiment consists of one AP and two dongles, which are plugged to STAs through USB-C cables. Additionally, the AP is powered by *Power over Ethernet (PoE)* and we plugged it

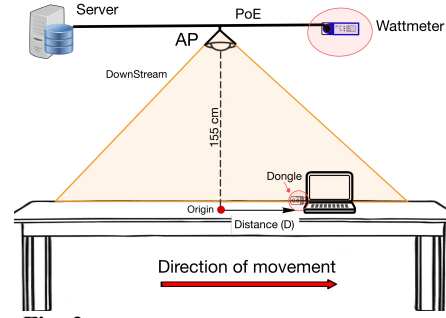


Fig. 2: Illustration of the experimental setup.

to a server machine. This product is mostly designed for indoor networking usage and can provide 100 Mbps transmission rate for both downlink and uplink using the infrared spectrum. OledComm follows the ITU G.vlc standard for AP [18], which employs the TDMA mechanism for the MAC layer. The LiFiMax AP can serve up to 16 STAs.

B. Experimental testbed

We conduct our experiments in an empty room to avoid introducing anomalies. We use the open-source tool *Iperf* [19] on the server to generate UDP downstream traffic at the application layer. We saturate the network (where the AP buffers are filled, allowing packet dropping) to determine the throughput limitations. We repeat each test at least 10 times to average the results obtained.

C. Network-wise performance parameters

First, we test the impact of the vertical distance between the AP and the STA. We observe that the throughput achieves its maximum when the distance between AP and STA is between 1 and 3 meters. We also study the impact of the reception angle, determined by distance, on throughput. For the LiFi STAs, we use dongles equipped with photodiodes that have a limited Field of View (FoV), restricting the angles at which the dongle can effectively receive optical signals. As a result, the position of the STA within the cell (i.e., the area covered by a LiFi AP) becomes an important factor. As shown in Figure 2, we move the STA horizontally away from the origin, where the reception angle is ideal (directly under the AP). As distance increases, the signal weakens and throughput drops because the limited FoV of the STA loses alignment with the AP. We measure the throughput at various distances from the origin and report the results in Table I. Note that the vertical distance between the AP and the STA is fixed at 155 cm. We observe in Table I that throughput drops significantly beyond 130cm, indicating the critical importance of STA placement. Therefore, to correctly serve a STA we consider that the maximum range for the LiFiMax is 150cm.

We also examine the behavior of the MAC layer by observing the measured throughput when two STAs compete to access the medium. To do so, we install a fixed STA (sta_1) under the AP, and a second STA (sta_2) which we gradually move away from sta_1 and thus degrade its performance. The results show that the degradation of the measured throughput of sta_2 does not affect the measured throughput of sta_1 (unless sta_2 becomes completely unreachable). This behavior is due to the TDMA mechanism, in which the AP shares the medium equally between its assigned STAs.

Distance (D in cm)		and	Throughput ($r_{ap,sta}$ in Mbps)		
D	$r_{ap,sta}$	D	$r_{ap,sta}$	D	$r_{ap,sta}$
0	94.2	60	94.08	120	87.91
5	94.2	70	94.12	130	86.96
10	94.0	80	94.04	140	68.49
25	94.11	90	94.22	150	34.24
40	94.14	100	94.13	155	0.0361
50	94.14	110	90.86		

TABLE I: Influence of reception angle on the STA throughput in a LiFi network.

D. Power consumption parameters

We measure the power consumption of the AP using a wattmeter, the *Voltcraft Energy-Logger 4000F*. Furthermore, we measure the power of the dongle plugged into the STA using a USB power meter, the *Type-C Tester S27*. The corresponding results are presented in Table II. We denote the power consumption of the AP as ρ_{ap} and that of the STA (dongle) as ρ_{sta} . A confidence interval 95% is represented using $\pm$.

Operational Mode	Transmission Rate (Mbps)	ρ_{ap} / ρ_{sta} (Watts)
AP		
Idle	N/A	Avg = 5.39 $\pm$ 0.02
Reception	10, 100, 10000	Avg = 6.48 $\pm$ 0.26
Transmission	10, 100, 10000	Avg = 11.35 $\pm$ 0.27
LiFi dongle (STA)		
Idle	N/A	Avg = 1.84 $\pm$ 0.08
Reception	10, 100, 10000	Avg = 2.03 $\pm$ 0.04
Transmission	10, 100, 10000	Avg = 2.76 $\pm$ 0.04

TABLE II: Power consumption of the LiFi AP and dongle (plugged to a STA).

As expected, both devices (AP and dongle) consume additional power in reception mode than in idle mode, since they convert the light signal into an electric signal. Furthermore, the transmission mode is the most power-consuming mode as it also requires more power for data transmission.

IV. ANALYTICAL FRAMEWORK

In this section, we introduce the framework used to evaluate the performance of each network deployment. We consider a HLWNet with n STAs and m APs (be it Wi-Fi or LiFi). We formulate the APA problem as an optimization problem aimed at maximizing network throughput and minimizing overall power consumption.

A. Modeling assumption

We represent the assignments between STAs and APs by an assignment matrix $\mathbf{Y} = [y_{ap,sta}]_{1 \leq ap \leq m; 1 \leq sta \leq n}$ of dimension $m \times n$. $y_{ap,sta}$ equals 1 only if sta is assigned to ap and equals 0 otherwise.

For the sake of simplicity, we made several assumptions: (i) We consider solely the downstream traffic, as it is the dominant traffic [20]; (ii) APs can be turned on or off, and when turned on, they can operate in transmission or idle mode; (iii) We adopt CSMA/CA for the Wi-Fi MAC layer and TDMA for the LiFi MAC layer; (iv) We neglect the potential interference between devices of the same technology (e.g., two LiFi APs). For Wi-Fi, the number of available channels allows nearby APs to operate on different channels (the 5GHz Wi-Fi has 24 non-overlapping channels). For LiFi, the interference can be mitigated by physical obstacles (e.g., walls).

B. Performance modeling

We denote by AP_w the set of Wi-Fi APs, δ_{sta} the throughput demand of a given STA sta , and $r_{ap,sta}$ the link capacity between sta and a given AP ap . We formulate the obtained throughput $d_{ap,sta}$ by a STA as: $\forall ap \in AP_w$

$$d_{ap,sta} = \min \left(\frac{\delta_{sta}}{\sum_{sta \in STAs} \frac{\delta_{sta} \times y_{ap,sta}}{r_{ap,sta}}}, \delta_{sta} \right) \times y_{ap,sta} \quad (1)$$

For LiFi, the obtained throughput is formulated to match the measurement results presented in Table I. We denote by AP_l the set of LiFi APs. We use a Lagrange polynomial approximation from the measurements presented in Section III to compute the link capacity as function of the distance between a STA and its LiFi AP. We consider the TDMA mechanism for LiFi, therefore, we divide $r_{ap,sta}$ by the total number of STAs assigned with ap . We formulate the obtained throughput $d_{ap,sta}$ of a STA as: $\forall ap \in AP_l$

$$d_{ap,sta} = \min \left(\frac{r_{ap,sta}}{\sum_{sta \in STAs} y_{ap,sta}}, \delta_{sta} \right) \times y_{ap,sta} \quad (2)$$

C. Power consumption modeling

As we consider only downstream traffic, the devices can only be in two operational modes: transmission or idle modes for the AP, and reception or idle modes for the STA. We formulate the total power consumption of APs and STAs as [7]:

$$P_{ap} = (1 - \tau_{uti,ap}) \times \rho_{id,ap} + \tau_{uti,ap} \times \rho_{tx,ap} \quad (3)$$

$$P_{sta} = (1 - \tau_{uti,sta}) \times \rho_{id,sta} + \tau_{uti,sta} \times \rho_{rx,sta} \quad (4)$$

where, P_{ap} and P_{sta} represent the total power consumed by ap and sta , respectively. $\tau_{uti,ap}$ (resp. $\tau_{uti,sta}$) is the utilization rate of ap (resp. sta). Finally, we use ρ_{tx} , ρ_{rx} , ρ_{id} to denote the power consumed by a device in transmission, reception, or idle modes, respectively. We assume that we can turn off the unassigned Wi-Fi and LiFi APs (i.e., APs that have no STAs assigned to them). In this case, the power consumption of these APs is equal to zero.

D. Optimization problem

The objective function aims to maximize the throughput of STA and minimize the overall power consumption (for both, APs and STAs). We express it as :

$$\text{maximize } F(\Phi, \Psi) \quad (5)$$

where Ψ is the total network power consumption (power consumed by every STA and every AP), and Φ is the throughput metric that accounts for the fairness between the different STAs. We formulate Φ as :

$$\Phi = \left(\prod_{\substack{ap \in APs, sta \in STAs : \\ y_{ap,sta} = 1}} \frac{d_{ap,sta}}{\delta_{sta}} \right)^{1/n} \quad (6)$$

where the ratio $\frac{d_{ap,sta}}{\delta_{sta}}$ in Φ measures the degree of satisfaction for each STA. We define our objective function F in Equation 7. We use Φ_{max} (resp. Φ_{min}) to denote the maximum (resp. minimum) STAs satisfaction. Analogously, we denote by Ψ_{max} (resp. Ψ_{min}) the maximum power consumption (resp. minimum).

$$F(\Phi, \Psi) = \left(\frac{\Phi - \Phi_{min}}{\Phi_{max} - \Phi_{min}} \right) \times \alpha + \left(\frac{\Psi_{max} - \Psi}{\Psi_{max} - \Psi_{min}} \right) \times (1 - \alpha) \quad (7)$$

We use α (resp. $1-\alpha$) to denote the weights assigned to the throughput metric Φ (resp. the consumption metric Ψ).

E. Heuristic

To solve the optimization problem, we employ a local search-based heuristic to find a feasible solution. Starting from a predetermined assignment matrix $\mathbf{Y}$, the heuristic advances toward its final solution by exploring the neighborhood of the current assignment matrix. We define this neighborhood as the set of matrices that can be obtained by making a single change to the assignment between a STA and an AP. At each iteration, the heuristic browses every STA. For each of them, it computes the resulting score from $F(\Phi, \Psi)$ to determine the best AP for each STA. The current iteration ends with the update of the assignment matrix so that the best-explored neighborhood assignment becomes the new current assignment. The process repeats until convergence is found (no changes improve the score of the objective function). To mitigate the odds of getting stuck in a local optimum, we repeat R times the local search with different random initial assignments (in our simulation, we consider $R=15$). Then, we keep the assignment matrix with the highest score as the best solution to the APA problem.

V. EVALUATION APPROACH AND USE CASES

In this section, we use the framework defined previously to identify when LiFi helps to improve the performance of WLANs. Our approach involves comparing different network deployments, where the framework determines the adequate number of APs for each technology, taking into account both throughput satisfaction and overall power consumption. In addition, we present the two use cases we use to illustrate the benefits of HLWNet in realistic contexts.

a) **Performance Evaluation:** The evaluation phase consists of four steps: (i) Determine the different deployments; (ii) Define the operational parameters (STAs, traffic, power consumption models, etc.); (iii) Find the best assignment matrix for each deployment using our heuristic and calculate the resulting score from the objective function (Eq. 7); (iv) Compare the scores to select the best deployment.

b) **LiFi consumption model:** To account for the potential diversity in the existing LiFi products, we consider two setups for the LiFi consumption model: (i) **Light-free power model:** The power consumption parameters are driven from our LiFiMax experimentation (Section III), where LiFi uses infrared, and do not piggyback on room lighting. (ii) **Light-assisted power model:** In this setup, LiFi uses the available visible light for data transmission. The LiFi APs do not consume power in idle mode and the transmission power is reduced by 50%, as supported by [21].

A. Use case 1: Enterprise setup

The 600 m^2 use case is an indoor downlink scenario based on an enterprise setup from IEEE [22]. The environment has 24 cubicles that form offices and a meeting room, as shown in Figure 3. Each office has four cubicles with partitions that reduce interference between the LiFi APs.

Deployment Design: We begin by determining the number and placement of Wi-Fi APs needed for full coverage. We explore deployments with 2 to 26 Wi-Fi APs. LiFi APs are placed where users (and hence STAs) are sedentary, like offices and meeting

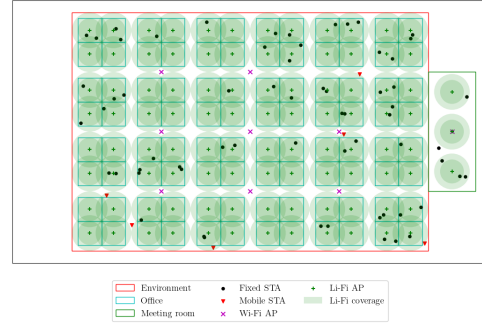


Fig. 3: Enterprise scenario with 4 LiFi APs per cube and a total of 9 Wi-Fi APs.

rooms, as shown in Figure 3. The meeting room has three LiFi APs and one Wi-Fi AP. By varying the number and location of APs from the two technologies, we obtain several different deployments. In total, we consider 70 different deployments.

Operational parameters: We consider two scenarios: a sparse scenario with 70 STAs and an ultra-dense scenario with 380 STAs. We consider 10% of the STAs to be mobile. The remaining 90% are uniformly distributed in the cubicles and meeting room. A cubicle holds a maximum of 4 STAs. We consider the two LiFi power models.

B. Use case 2: Cisco Meraki office

This use case represents a highly-dense scenario with a Wi-Fi deployment. It is inspired from the 4th floor of the Cisco Meraki office in San Francisco that covers 5505 m^2 , as shown in Figure 4. The Wi-Fi APs deployment is the one of Cisco Meraki. This enables us to assess the potential benefits of a LiFi deployment on an existing and well-sized Wi-Fi network.

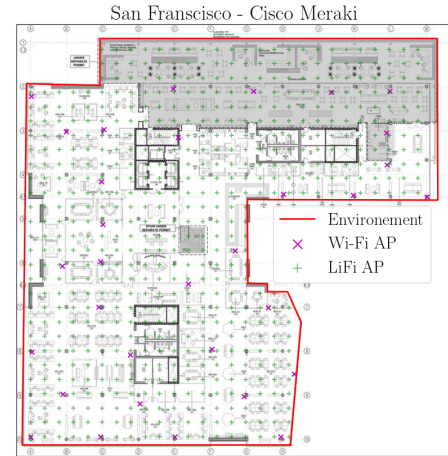


Fig. 4: High density scenario, Cisco San Francisco, 4th Floor.

Deployment Design: Since the Wi-Fi deployment is already fixed by the choices made by Cisco Meraki, we only need to place LiFi APs. We cover the entire area with LiFi APs, spaced every three meters, as shown in Figure 4.

Operational parameters: For this use case, we vary the number of STAs from 2 to 450. We assume 10% of the STAs are mobile and are assigned to Wi-Fi APs, while the other 90% are uniformly distributed across the environment, not limited to the office. Only the light-assisted power model is considered in this use case.

VI. NUMERICAL RESULTS

We implement our simulator with the performance models and use cases of Section V in Python. We present the simulation parameters in Table III. Each deployment is evaluated at least 10 times to average the results. Furthermore, certain figures include the values corresponding to the first and third quartiles.

Parameter	Value		
	Traffic		
Transport protocol	UDP		
Packet size	1440 Bytes		
δ_{sta}	[10–20] Mbps		
	AP	Wi-Fi	LiFi
Standard		802.11ax	ITU-T G.vlc
Range (m)		51m	1.5m
$r_{ap,sta}$		[154–210] Mbps	[0.03-94.2] Mbps
$\rho_{id,ap}$		13.1 Watt	5.4 Watt
$\rho_{tx,ap}$		18.24 Watt	11.35 Watt
	STA	Wi-Fi	LiFi
$\rho_{id,sta}$		1.4 Watt	1.84 Watt
$\rho_{rx,sta}$		1.6 Watt	2.04 Watt

TABLE III: Simulation parameters.

A. Use case 1: Enterprise setup

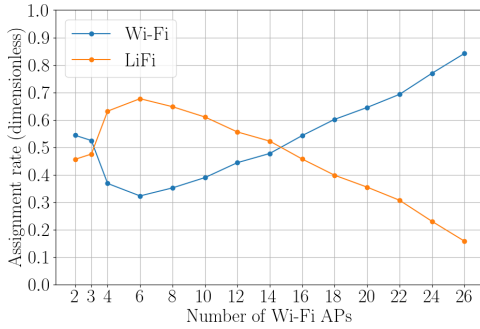


Fig. 5: Use case 1 : Rate of STAs assigned to LiFi APs and Wi-Fi APs for the 4 LiFi per cube deployment.

Light-free power model: Figures 5 and 6 show the numerical results using the light-free power model with five LiFi deployments (0 to 4 LiFi APs per cube) and fourteen Wi-Fi deployments (2 to 26 Wi-Fi APs) with 380 STAs. Figure 6(a) shows the overall throughput satisfaction (metric Φ). We observe that 100% satisfaction is achieved only with 3 LiFi per cube and 26 Wi-Fi APs, or with 4 LiFi per cube and at least 16 Wi-Fi APs. Other LiFi deployments max out at a satisfaction rate of 95%.

Figure 6(b) shows the overall power consumption of the network for different deployments (metric Ψ). The deployment with **0 LiFi per cube** uses the least power consumption, while **the 4 LiFi per cube** setup uses **twice as much power** with 10 Wi-Fi APs. When enough Wi-Fi APs are available to handle traffic, our heuristic assigns STAs to Wi-Fi and turns off LiFi APs to reduce power consumption. This is shown in Figure 5, which illustrates the proportion of STAs assigned to Wi-Fi and LiFi in the 4 LiFi per cube deployment. With 3 to 14 Wi-Fi APs, most STAs are assigned to LiFi. However, starting from 6 Wi-Fi APs, our heuristic switches STAs assignments to Wi-Fi, allowing some LiFi APs to turn off, and decreasing its power consumption from 6 Wi-Fi APs, as shown in Figure 6(b). Figure 6(c) shows the scores of our objective function (Eq. 7). The scores range from

0.3 to 0.73. The best score, 0.73, comes from the 0 LiFi APs per cube and only the Wi-Fi deployment. This deployment achieves up to 95% throughput satisfaction while using the least power.

Light-assisted power model: The results of the light-assisted power model are not shown here. However, with this model (which favors LiFi APs as their power consumption is lower), the score of the 4 LiFi per cube deployment reaches a stable plateau starting from 8 Wi-Fi APs and receives the highest value (0.8). Indeed, since LiFi consumes less power, the proportion of STAs assigned to LiFi APs is higher than that assigned to Wi-Fi APs.

B. Use case 2: Cisco Meraki office

We present the performance results for the Cisco Meraki use case in Figure 7. In Figure 7(a), we observe that as the number of STAs increases, the network increasingly relies on LiFi and Wi-Fi technologies, hence, the LiFi APs assist the network by handling the STAs. However, even when considering the *light-assisted model*, the Wi-Fi APs dominate the STAs assignments. This can be seen in Figure 7(b), where the network uses 85% of the Wi-Fi APs and almost 30% of the LiFi APs. Starting from 400 STAs, we can observe that to cover almost the same proportion of STAs, we need on average 27 Wi-Fi APs and **6 times more** LiFi APs as shown in Figure 7(c).

VII. CONCLUSIONS AND FUTURE WORK

In this paper, we investigate the efficiency of hybrid LiFi/Wi-Fi networks for WLANs in terms of their ability to meet the QoS of STAs while reducing the overall power consumption of WLANs. Our study presents two power consumption models, one of which is derived from our experimentation with an off-the-shelf LiFi product. In addition, we introduce an approach based on the APA problem to find adequate WLAN deployments.

Our results show that finding the right balance between LiFi and Wi-Fi, in terms of deployment or assignment, is strongly influenced by the power consumption of LiFi, the placement of STAs (and the LiFi ability to pool several STAs) and traffic demand. In other words, this balance is highly scenario-dependent and depends on the particular equipment used. Therefore, a preliminary assessment like ours is a necessary step in the deployment of a high-performance HLWNet network.

Considering real-world use cases, our numerical results show that Wi-Fi dominates the WLANs, especially when Wi-Fi deployment ensures good coverage and throughput. The HLWNets are advantageous for WLAN only if LiFi consumes significantly less power than Wi-Fi (typically by piggybacking transmissions with lighting) and if it is capable of pooling several STAs (if several STAs can assign to the same LiFi AP). Otherwise, the higher radio range of Wi-Fi and the ability of its APs to handle more STAs make it a better solution both in terms of throughput and power consumption.

As future work, we plan to further analyze the environmental impact of LiFi devices. This includes conducting a life cycle assessment of LiFi. Performing such a deep analysis can help uncover, or assess, the real environmental costs associated with LiFi.

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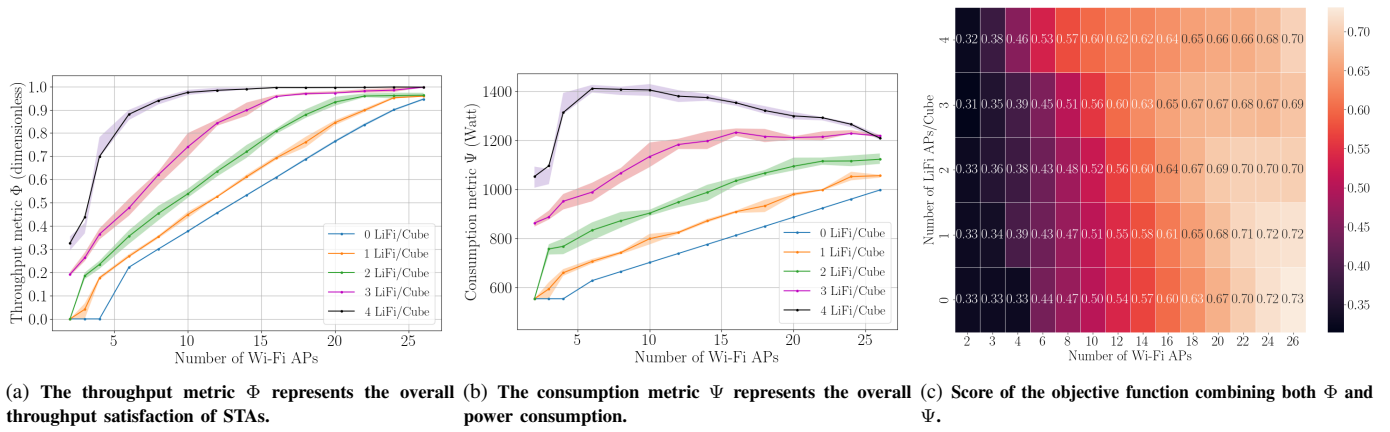


Fig. 6: Enterprise setup: performance results for different deployments, considering Light-free power model.

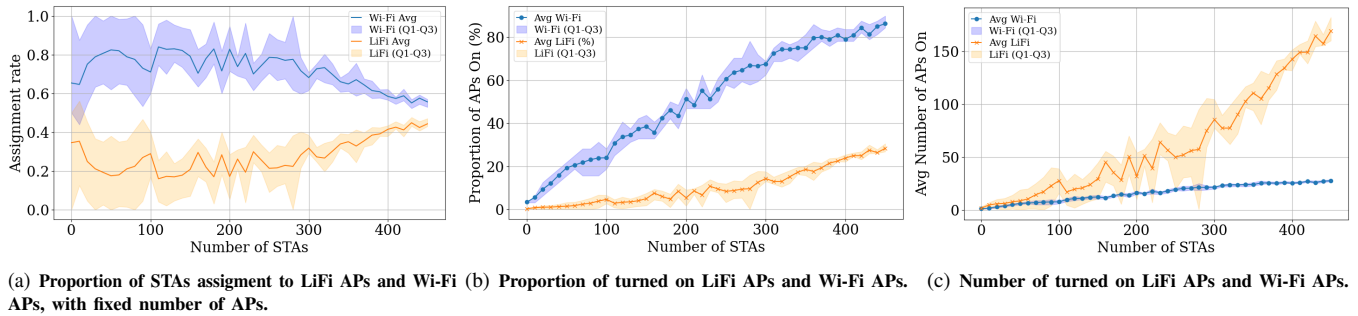


Fig. 7: Cisco Meraki deployment: performance results with fixed number of APs, considering Light-assisted power model.

REFERENCES

- [1] M. Obeed, A. M. Salhab, M.-S. Alouini, and S. A. Zummo, "On optimizing vlc networks for downlink multi-user transmission: A survey," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2947–2976, 2019.
- [2] M. Z. Chowdhury, M. K. Hasan, M. Shahjalal, M. T. Hossain, and Y. M. Jang, "Optical wireless hybrid networks: Trends, opportunities, challenges, and research directions," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 2, pp. 930–966, 2020.
- [3] X. Wu, M. D. Soltani, L. Zhou, M. Safari, and H. Haas, "Hybrid lifi and wifi networks: A survey," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 1398–1420, 2021.
- [4] A. E. Ibhahe, A. L. Imoize, and O. Okoyeigbo, "A brief overview of energy efficiency resources in emerging wireless communication systems," *Telecom*, vol. 3, no. 2, pp. 281–300, 2022. [Online]. Available: <https://www.mdpi.com/2673-4001/3/2/16>
- [5] X. Wu and H. Haas, "Access point assignment in hybrid lifi and wifi networks in consideration of lifi channel blockage," in *2017 IEEE 18th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2017, pp. 1–5.
- [6] X. Wu, M. Safari, and H. Haas, "Access point selection for hybrid li-fi and wi-fi networks," *IEEE Transactions on Communications*, vol. 65, no. 12, pp. 5375–5385, 2017.
- [7] E. Guérin, T. Begin, A. Busson, and I. Guérin-Lassous, "Towards a Throughput and Energy Efficient Association Strategy for Wi-Fi/LiFi Heterogeneous Networks," in *PE-WASUN 2021 - 18th ACM International Symposium on Performance Evaluation of Wireless Ad Hoc, Sensor, and Ubiquitous Networks*. Alicante, Spain: ACM, Nov. 2021. [Online]. Available: <https://hal.science/hal-03445734>
- [8] X. Li, R. Zhang, and L. Hanzo, "Cooperative load balancing in hybrid visible light communications and wifi," *IEEE Transactions on Communications*, vol. 63, no. 4, pp. 1319–1329, 2015.
- [9] X. Wu, M. Safari, and H. Haas, "Joint optimisation of load balancing and handover for hybrid lifi and wifi networks," in *2017 IEEE Wireless Communications and Networking Conference (WCNC)*, 2017, pp. 1–5.
- [10] Y. Wang, X. Wu, and H. Haas, "Load balancing game with shadowing effect for indoor hybrid lifi/rf networks," *IEEE Transactions on Wireless Communications*, vol. 16, no. 4, pp. 2366–2378, 2017.
- [11] X. Wu and H. Haas, "Load balancing for hybrid lifi and wifi networks: To tackle user mobility and light-path blockage," *IEEE Transactions on Communications*, vol. 68, no. 3, pp. 1675–1683, 2020.
- [12] M. Obeed, A. M. Salhab, S. A. Zummo, and M.-S. Alouini, "Joint optimization of power allocation and load balancing for hybrid vlc/rf networks," *Journal of Optical Communications and Networking*, vol. 10, no. 5, pp. 553–562, 2018.
- [13] H. Vijayaraghavan and W. Kellerer, "Delay-aware wireless resource allocation and user association in lifi-wifi heterogeneous networks," in *2021 IEEE Global Communications Conference (GLOBECOM)*, 2021, pp. 01–06.
- [14] Y. Wang, X. Wu, and H. Haas, "Fuzzy logic based dynamic handover scheme for indoor li-fi and rf hybrid network," in *2016 IEEE International Conference on Communications (ICC)*, 2016, pp. 1–6.
- [15] X. Wu, D. C. O'Brien, X. Deng, and J.-P. M. G. Linnartz, "Smart handover for hybrid lifi and wifi networks," *IEEE Transactions on Wireless Communications*, vol. 19, no. 12, pp. 8211–8219, 2020.
- [16] D. Tsonev, S. Videv, and H. Haas, "Towards a 100 gb/s visible light wireless access network," *Opt. Express*, vol. 23, no. 2, pp. 1627–1637, Jan 2015. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?URI=oe-23-2-1627>
- [17] M. S. Islim and h. haas, "Modulation techniques for li-fi," *ZTE COMMUNICATIONS*, vol. 14, pp. 29–40, 04 2016.
- [18] B. Béchadegue and B. Azoulay, "An industrial view on lifi challenges and future," in *2020 12th International Symposium on Communication Systems, Networks and Digital Signal Processing (CSNDSP)*, 2020, pp. 1–6.
- [19] L. B. N. L. ESnet, "Iperf3: A TCP, UDP, and SCTP network bandwidth measurement tool," <https://github.com/esnet/iperf>, Nov. 2021.
- [20] A. Gupta, J. Min, and I. Rhee, "Wifox: Scaling wifi performance for large audience environments," in *Proceedings of the 8th International Conference on Emerging Networking Experiments and Technologies*, ser. CoNEXT '12. New York, NY, USA: Association for Computing Machinery, 2012, pp. 217–228. [Online]. Available: <https://doi.org/10.1145/2413176.2413202>
- [21] M. Kashef, M. Ismail, M. Abdallah, K. A. Qaraqe, and E. Serpedin, "Energy efficient resource allocation for mixed rf/vlc heterogeneous wireless networks," *IEEE Journal on Selected Areas in Communications*, vol. 34, no. 4, pp. 883–893, 2016.
- [22] H. S. Simone Merlin, Gwen Barriac and al, *Gax simulation scenarios*. doc. IEEE 802.11-14/0980r16, 2015.